Newman-Penrose Formalism Calculations

Stephani et al., Equation (37.51_2)

Metric

$$ds^2 = -\mathrm{d}t \otimes \mathrm{d}t + \left(-\frac{F\left(t\right)^2}{r^2-1}\right) \mathrm{d}r \otimes \mathrm{d}r + V\left(t,z,r,\phi\right)^2 \mathrm{d}z \otimes \mathrm{d}z + r^2 F\left(t\right)^2 \mathrm{d}\phi \otimes \mathrm{d}\phi$$

Coordinates

$$\left(x^\mu\right) = \left(t, r, z, \phi\right)$$

Null Tetrad

Covariant components

$$l = \left(\sqrt{\frac{1}{2}}\right) dt + \left(\frac{\sqrt{2}F\left(t\right)}{2\sqrt{-r^2+1}}\right) dr$$

$$n = \left(\sqrt{\frac{1}{2}}\right) dt + \left(-\frac{\sqrt{2}F\left(t\right)}{2\sqrt{-r^2+1}}\right) dr$$

$$m = \left(\frac{1}{2}\sqrt{2}V\left(t,z,r,\phi\right)\right) dz + \left(\frac{1}{2}i\sqrt{2}rF\left(t\right)\right) d\phi$$

$$\overline{m} = \left(\frac{1}{2}\sqrt{2}V\left(t,z,r,\phi\right)\right) dz + \left(-\frac{1}{2}i\sqrt{2}rF\left(t\right)\right) d\phi$$

Contravariant components

$$l = \left(-\frac{1}{2}\sqrt{2}\right) dt + \left(\frac{\sqrt{2}\sqrt{|r+1|}\sqrt{|r-1|}}{2\,F\left(t\right)}\right) dr$$

$$n = \left(-\frac{1}{2}\,\sqrt{2}\right)\,dt + \left(\frac{\sqrt{2}(r+1)(r-1)}{2\,F\left(t\right)\,\sqrt{|r+1|}\sqrt{|r-1|}}\right)\,dr$$

$$m = \left(\frac{\sqrt{2}}{2\,V\left(t,z,r,\phi\right)}\right)\,dz + \left(\frac{i\,\sqrt{2}}{2\,rF\left(t\right)}\right)\,d\phi$$

$$\overline{m} = \left(\frac{\sqrt{2}}{2\,V\left(t,z,r,\phi\right)}\right)\,dz + \left(-\frac{i\,\sqrt{2}}{2\,rF\left(t\right)}\right)\,d\phi$$

Directional Derivatives

$$D = -\frac{1}{2}\,\sqrt{2}\,\partial_t + \frac{\sqrt{2}\sqrt{|r+1|}\sqrt{|r-1|}}{2\,F\left(t\right)}\,\partial_r$$

$$\Delta = -\frac{1}{2}\,\sqrt{2}\,\partial_t + \frac{\sqrt{2}(r+1)(r-1)}{2\,F\left(t\right)\,\sqrt{|r+1|}\sqrt{|r-1|}}\,\partial_r$$

$$\delta = \frac{\sqrt{2}}{2\,V\left(t,z,r,\phi\right)}\,\partial_z + \frac{i\,\sqrt{2}}{2\,rF\left(t\right)}\,\partial_\phi$$

$$\overline{\delta} = \frac{\sqrt{2}}{2\,V\left(t,z,r,\phi\right)}\,\partial_z - \frac{i\,\sqrt{2}}{2\,rF\left(t\right)}\,\partial_\phi$$

Spin Coefficients

$$\rho = \frac{\sqrt{2}\Big(rV\left(t,z,r,\phi\right)\,\sqrt{|r+1|}\sqrt{|r-1|}\,\frac{\partial}{\partial t}F\left(t\right) + rF\left(t\right)\,\sqrt{|r+1|}\sqrt{|r-1|}\,\frac{\partial}{\partial t}V\left(t,z,r,\phi\right) + r^3\,\frac{\partial}{\partial r}V\left(t,z,r,\phi\right) + r^2V\left(t,z,r,\phi\right) - r\,\frac{\partial}{\partial r}V\left(t,z,r,\phi\right) - V\left(t,z,r,\phi\right)\Big)}{4\,rF\left(t\right)V\left(t,z,r,\phi\right)\,\sqrt{|r+1|}\sqrt{|r-1|}}$$

$$\sigma = -\frac{\sqrt{2}\Big(rV\left(t,z,r,\phi\right)\,\sqrt{|r+1|}\sqrt{|r-1|}\,\frac{\partial}{\partial t}F\left(t\right) - rF\left(t\right)\,\sqrt{|r+1|}\sqrt{|r-1|}\,\frac{\partial}{\partial t}V\left(t,z,r,\phi\right) - r^3\,\frac{\partial}{\partial r}V\left(t,z,r,\phi\right) + r^2V\left(t,z,r,\phi\right) + r\,\frac{\partial}{\partial r}V\left(t,z,r,\phi\right) - V\left(t,z,r,\phi\right)\Big)}{4\,rF\left(t\right)V\left(t,z,r,\phi\right)\,\sqrt{|r+1|}\sqrt{|r-1|}}$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}\Big(rV\left(t,z,r,\phi\right)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F\left(t\right) + rF\left(t\right)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}V\left(t,z,r,\phi\right) - r^{3}\frac{\partial}{\partial r}V\left(t,z,r,\phi\right) - r^{2}V\left(t,z,r,\phi\right) + r\frac{\partial}{\partial r}V\left(t,z,r,\phi\right) + V\left(t,z,r,\phi\right)\Big)}{4\,rF\left(t\right)V\left(t,z,r,\phi\right)\sqrt{|r+1|}\sqrt{|r-1|}}$$

$$\lambda = \frac{\sqrt{2}\Big(rV\left(t,z,r,\phi\right)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F\left(t\right) - rF\left(t\right)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}V\left(t,z,r,\phi\right) + r^{3}\frac{\partial}{\partial r}V\left(t,z,r,\phi\right) - r^{2}V\left(t,z,r,\phi\right) - r\frac{\partial}{\partial r}V\left(t,z,r,\phi\right) + V\left(t,z,r,\phi\right)\Big)}{4\,rF\left(t\right)V\left(t,z,r,\phi\right)\sqrt{|r+1|}\sqrt{|r-1|}}$$

$$\nu = 0$$

$$\alpha = \frac{i\,\sqrt{2}\frac{\partial}{\partial\phi}V\left(t,z,r,\phi\right)}{4\,rF\left(t\right)V\left(t,z,r,\phi\right)}$$

$$\beta = \frac{i\,\sqrt{2}\frac{\partial}{\partial\phi}V\left(t,z,r,\phi\right)}{4\,rF\left(t\right)V\left(t,z,r,\phi\right)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{\sqrt{2}\frac{\partial}{\partial t}F\left(t\right)}{4\,F\left(t\right)}$$

$$\gamma = -\frac{\sqrt{2}\left(r^{2} - |r+1||r-1| - 1\right)\frac{\partial}{\partial t}F\left(t\right)}{8\,F\left(t\right)|r+1||r-1|}$$

Ricci Tensor Components

$$\Phi_{00} = \frac{r^4 V(t,z,r,\phi) \frac{\partial}{\partial t} F(t)^2 - r^2 V(t,z,r,\phi) |r+1||r-1| \frac{\partial}{\partial t} F(t)^2 - 2\, r^4 F(t) V(t,z,r,\phi) \frac{\partial^2}{(\partial t)^2} F(t) + 2\, r^4 F(t) \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t,z,r,\phi) - r^4 F(t)^2 \frac{\partial^2}{(\partial t)^2} V(t,z,r,\phi) + r^2 F(t)^2 |r+1||r-1| \frac{\partial^2}{(\partial t)^2}}{}$$

$$\Phi_{01} = \frac{i\, r F(t) \frac{\partial^2}{\partial t \partial \phi} V(t,z,r,\phi) - i\, r \sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial^2}{\partial r \partial \phi} V(t,z,r,\phi) - i\, r \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial \phi} V(t,z,r,\phi) + i \sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial}{\partial \phi} V(t,z,r,\phi)}{4\, r^2 F(t)^2 V(t,z,r,\phi)}$$

$$\Phi_{02} = -\frac{V(t,z,r,\phi) \frac{\partial}{\partial t} F(t)^2 + F(t) V(t,z,r,\phi) \frac{\partial^2}{(\partial t)^2} F(t) - F(t) \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t,z,r,\phi) - F(t)^2 \frac{\partial^2}{(\partial t)^2} V(t,z,r,\phi) - r^2 \frac{\partial^2}{(\partial r)^2} V(t,z,r,\phi) - r \frac{\partial}{\partial r} V(t,z,r,\phi) + V(t,z,r,\phi) + \frac{\partial^2}{(\partial r)^2} V(t,z,r,\phi)}{4\, F(t)^2 V(t,z,r,\phi)}$$

$$\Phi_{10} = -\frac{i\, r F(t) \frac{\partial^2}{\partial t \partial \phi} V(t,z,r,\phi) - i\, r \sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial^2}{\partial r \partial \phi} V(t,z,r,\phi) - i\, r \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial \phi} V(t,z,r,\phi) + i \sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial}{\partial \phi} V(t,z,r,\phi)}{4\, r^2 F(t)^2 V(t,z,r,\phi)}$$

$$\Phi_{11} = -\frac{2\, r^2 F(t) V(t,z,r,\phi) \sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial^2}{(\partial t)^2} F(t) - 2\, r^2 F(t) \sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t,z,r,\phi) - r^4 F(t) \frac{\partial^2}{\partial t \partial r} V(t,z,r,\phi) - r^2 F(t) |r+1||r-1| \frac{\partial^2}{\partial t \partial r} V(t,z,r,\phi) + r^4 \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial r} V}{8\, r}$$

$$\Phi_{12} = \frac{i\, r F(t) \sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial^2}{\partial t \partial \phi} V(t,z,r,\phi) - i\, r^3 \frac{\partial^2}{\partial r \partial \phi} V(t,z,r,\phi) - i\, r \sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial \phi} V(t,z,r,\phi) + i\, r^2 \frac{\partial}{\partial \phi} V(t,z,r,\phi) + i\, r \frac{\partial^2}{\partial r \partial \phi} V(t,z,r,\phi) - i \frac{\partial}{\partial \phi} V(t,z,r,\phi)}{4\, r^2 F(t)^2 V(t,z,r,\phi) \sqrt{|r+1|}\sqrt{|r-1|}}$$

$$\Phi_{20} = -\frac{V(t,z,r,\phi) \frac{\partial}{\partial t} F(t)^2 + F(t) V(t,z,r,\phi) \frac{\partial^2}{(\partial t)^2} F(t) - F(t) \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t,z,r,\phi) - F(t)^2 \frac{\partial^2}{(\partial t)^2} V(t,z,r,\phi) - r^2 \frac{\partial^2}{(\partial r)^2} V(t,z,r,\phi) - r \frac{\partial}{\partial r} V(t,z,r,\phi) + V(t,z,r,\phi) + \frac{\partial^2}{(\partial r)^2} V(t,z,r,\phi)}{4\, F(t)^2 V(t,z,r,\phi)}$$

$$\Phi_{21} = -\frac{i\, r F(t) \sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial^2}{\partial t \partial \phi} V(t,z,r,\phi) - i\, r^3 \frac{\partial^2}{\partial r \partial \phi} V(t,z,r,\phi) - i\, r \sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial \phi} V(t,z,r,\phi) + i\, r^2 \frac{\partial}{\partial \phi} V(t,z,r,\phi) + i\, r \frac{\partial^2}{\partial r \partial \phi} V(t,z,r,\phi) - i \frac{\partial}{\partial \phi} V(t,z,r,\phi)}{4\, r^2 F(t)^2 V(t,z,r,\phi) \sqrt{|r+1|}\sqrt{|r-1|}}$$

$\Phi_{22} =$

$$-\frac{r^4 V(t,z,r,\phi)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F(t)^2 - r^2 V(t,z,r,\phi)|r+1|^{\frac{3}{2}}|r-1|^{\frac{3}{2}}\frac{\partial}{\partial t}F(t)^2 + 2r^2 F(t) V(t,z,r,\phi)|r+1|^{\frac{3}{2}}|r-1|^{\frac{3}{2}}\frac{\partial^2}{(\partial t)^2}F(t) - 2r^2 F(t)|r+1|^{\frac{3}{2}}|r-1|^{\frac{3}{2}}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial t}V(t,z,r,\phi) - }{}$$

Λ

$$= \frac{r^2 V(t,z,r,\phi)\frac{\partial}{\partial t}F(t)^2 + 2r^2 F(t) V(t,z,r,\phi)\frac{\partial^2}{(\partial t)^2}F(t) + 2r^2 F(t)\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial t}V(t,z,r,\phi) + r^2 F(t)^2\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi) + r^4\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi) + 2r^3\frac{\partial}{\partial r}V(t,z,r,\phi) + r^2 V(t,z,r,\phi) - r^2\frac{\partial^2}{(\partial r}}{12\, r^2 F(t)^2 V(t,z,r,\phi)}$$

Weyl Tensor Components

$\Psi_0 =$

$$-\frac{r^2 V(t,z,r,\phi)\frac{\partial}{\partial t}F(t)^2 - V(t,z,r,\phi)|r+1||r-1|\frac{\partial}{\partial t}F(t)^2 - r^2 F(t) V(t,z,r,\phi)\frac{\partial^2}{(\partial t)^2}F(t) + F(t) V(t,z,r,\phi)|r+1||r-1|\frac{\partial^2}{(\partial t)^2}F(t) - r^2 F(t)\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial t}V(t,z,r,\phi) + F(t)|r+1||r}{}$$

Ψ_1

$$= \frac{\left(rF(t)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi) + r^3\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi) - r\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial\phi}V(t,z,r,\phi) - r^2\frac{\partial}{\partial\phi}V(t,z,r,\phi) - r\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi) + \frac{\partial}{\partial\phi}V(t,z,r,\phi)\right)\left(-i\,r^2 + i\,|r+1||r\right.}{8\,(r+1)(r-1)r^2 F(t)^2 V(t,z,r,\phi)\sqrt{|r+1|}\sqrt{|r-1|}}$$

Ψ_2

$$= \frac{r^2 V(t,z,r,\phi)\frac{\partial}{\partial t}F(t)^2 - r^2 F(t) V(t,z,r,\phi)\frac{\partial^2}{(\partial t)^2}F(t) - r^2 F(t)\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial t}V(t,z,r,\phi) + r^2 F(t)^2\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi) + r^4\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi) - r^3\frac{\partial}{\partial r}V(t,z,r,\phi) + r^2 V(t,z,r,\phi) - r^2\frac{\partial^2}{(\partial r)^2}V(}{12\, r^2 F(t)^2 V(t,z,r,\phi)}$$

Ψ_3

$$= \frac{\left(rF(t)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi) - r^3\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi) - r\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial\phi}V(t,z,r,\phi) + r^2\frac{\partial}{\partial\phi}V(t,z,r,\phi) + r\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi) - \frac{\partial}{\partial\phi}V(t,z,r,\phi)\right)\left(-i\,r^2 + i\,|r+1||r\right.}{8\, r^2 F(t)^2 V(t,z,r,\phi)|r+1|^{\frac{3}{2}}|r-1|^{\frac{3}{2}}}$$

Ψ_4

$$= \frac{r^2 V(t,z,r,\phi)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F(t)^2 - V(t,z,r,\phi)|r+1|^{\frac{3}{2}}|r-1|^{\frac{3}{2}}\frac{\partial}{\partial t}F(t)^2 - r^2 F(t) V(t,z,r,\phi)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^2}{(\partial t)^2}F(t) + F(t) V(t,z,r,\phi)|r+1|^{\frac{3}{2}}|r-1|^{\frac{3}{2}}\frac{\partial^2}{(\partial t)^2}F(t) - r^2 F(t)}{}$$

Newman-Penrose Equations

$NP1 =$

$$-\frac{\left(r^2 V(t,z,r,\phi)^2 \sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F(t)^2 - r^2 F(t)^2 V(t,z,r,\phi)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi) - 2\,r^2 F(t)\,V(t,z,r,\phi)\,|r+1||r-1|\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi) + 4\,r^2 V(t,z,r,\phi)\,|r+1||r-\right.}{}$$

$NP2$

$$= \frac{\left(r^2 V(t,z,r,\phi)^2 \sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F(t)^2 - r^2 F(t)\,V(t,z,r,\phi)^2\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^2}{(\partial t)^2}F(t) - r^2 F(t)\,V(t,z,r,\phi)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial t}V(t,z,r,\phi) + r^2 F(t)^2 V(t,z,r,\phi)\sqrt{|r+\right.}{}$$

$NP3$

$$= \frac{\left(rF(t)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi) - r^3\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi) - r\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial\phi}V(t,z,r,\phi) + r^2\frac{\partial}{\partial\phi}V(t,z,r,\phi) + r\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi) - \frac{\partial}{\partial\phi}V(t,z,r,\phi)\right)\left(i\,r^2 + i\,|r+1||r-\right.}{8\,(r+1)(r-1)r^2F(t)^2\,V(t,z,r,\phi)\sqrt{|r+1|}\sqrt{|r-1|}}$$

$$NP4 = \frac{\left(i\,r^2 + i\,|r+1||r-1| - i\right)\frac{\partial}{\partial r}V(t,z,r,\phi)\,\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{4\,rF(t)^2\,V(t,z,r,\phi)^2\sqrt{|r+1|}\sqrt{|r-1|}}$$

$NP5$

$$= \frac{\left(rF(t)\,V(t,z,r,\phi)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi) - r^3 V(t,z,r,\phi)\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi) - rV(t,z,r,\phi)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial\phi}V(t,z,r,\phi) + 2\,r^3\frac{\partial}{\partial r}V(t,z,r,\phi)\frac{\partial}{\partial\phi}V(t,z,r,\phi) + r^2 V\right.}{8\,(r+1)(r-1)r^2F(t)^2\,V(t,z,r,\phi)^2\sqrt{|r+1}}$$

$NP6 =$

$$-\frac{\left(V(t,z,r,\phi)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F(t)^2 + F(t)\,V(t,z,r,\phi)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^2}{(\partial t)^2}F(t) - r^2F(t)\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi) + r^2\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial r}V(t,z,r,\phi) + F(t)\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi) - \frac{\partial}{\partial t}F(t)\frac{\partial}{\partial r}V(t,\right.}{8\,F(t)^2\,V(t,z,r,\phi)\,|r+1|^{\frac{3}{2}}|r-1|^{\frac{3}{2}}}$$

$NP7$

$$= \frac{\left(r^2F(t)\,V(t,z,r,\phi)\,|r+1||r-1|\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi) - r^2F(t)\,|r+1||r-1|\frac{\partial}{\partial t}V(t,z,r,\phi)\frac{\partial}{\partial r}V(t,z,r,\phi) + r^4\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial r}V(t,z,r,\phi)^2 - r^2V(t,z,r,\phi)^2\sqrt{|r+1|}\sqrt{|r-1|} - r^2\sqrt{|r\right.}{4\,r^2F(t)^2\,V(t,z,r,\phi)^2\,|r+1|^{\frac{3}{2}}|r-1|^{\frac{3}{2}}}$$

$$NP8 = \frac{\left(r^2 F(t)\, V(t,z,r,\phi)\, |r+1||r-1| \frac{\partial^2}{\partial t \partial r} V(t,z,r,\phi) - r^2 F(t)\, |r+1||r-1| \frac{\partial}{\partial t} V(t,z,r,\phi)\, \frac{\partial}{\partial r} V(t,z,r,\phi) + r^4 \sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial}{\partial r} V(t,z,r,\phi)^2 + r^2 V(t,z,r,\phi)^2 \sqrt{|r+1|}\sqrt{|r-1|} - r^2 \sqrt{|r}\right.}{4\, r^2 F(t)^2\, V(t,z,r,\phi)^2\, |r+1|^{\frac{3}{2}}\, |r-1|^{\frac{3}{2}}}$$

$$NP9 = -\frac{\left(rF(t)\, \frac{\partial^2}{\partial t \partial \phi} V(t,z,r,\phi) - r\sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial^2}{\partial r \partial \phi} V(t,z,r,\phi) - r \frac{\partial}{\partial t} F(t)\, \frac{\partial}{\partial \phi} V(t,z,r,\phi) + \sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial}{\partial \phi} V(t,z,r,\phi)\right)\left(i\, r^2 + i\, |r+1||r-1| - i\right)}{8\,(r+1)(r-1) r^2 F(t)^2\, V(t,z,r,\phi)}$$

$$NP10 = \frac{\left(rF(t)\, V(t,z,r,\phi)\, \sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial^2}{(\partial t)^2} F(t) - rF(t)^2 \sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial^2}{(\partial t)^2} V(t,z,r,\phi) + r|r+1||r-1| \frac{\partial}{\partial t} F(t)\, \frac{\partial}{\partial r} V(t,z,r,\phi) - r^3 \sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial^2}{(\partial r)^2} V(t,z,r,\phi) - V(t,z,r,\phi)\, |\right.}{8\, rF(t)^2\, V(t,z,r,\phi)\, |r+1|^{\frac{3}{2}} |r-}$$

$$NP11 = -\frac{\left(rF(t)\, \sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial^2}{\partial t \partial \phi} V(t,z,r,\phi) + 3\, r^3 \frac{\partial^2}{\partial r \partial \phi} V(t,z,r,\phi) - r\sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial}{\partial t} F(t)\, \frac{\partial}{\partial \phi} V(t,z,r,\phi) - 3\, r^2 \frac{\partial}{\partial \phi} V(t,z,r,\phi) - 3\, r \frac{\partial^2}{\partial r \partial \phi} V(t,z,r,\phi) + 3\, \frac{\partial}{\partial \phi} V(t,z,r,\phi)\right)\left(i\, r^2 + i\, |r\right.}{8\,(r+1)(r-1) r^2 F(t)^2\, V(t,z,r,\phi)\, \sqrt{|r+1|}\sqrt{|r-1|}}$$

$$NP12 = \frac{\left(rF(t)\, \frac{\partial^2}{\partial t \partial r} V(t,z,r,\phi) - r \frac{\partial}{\partial t} F(t)\, \frac{\partial}{\partial r} V(t,z,r,\phi) + 4\, \sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial}{\partial r} V(t,z,r,\phi)\right)\left(r^2 + |r+1||r-1| - 1\right)}{8\, rF(t)^2\, V(t,z,r,\phi)\, \sqrt{|r+1|}\sqrt{|r-1|}}$$

$$NP13 = \frac{\left(rF(t)\, \frac{\partial^2}{\partial t \partial \phi} V(t,z,r,\phi) - r\sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial^2}{\partial r \partial \phi} V(t,z,r,\phi) - r \frac{\partial}{\partial t} F(t)\, \frac{\partial}{\partial \phi} V(t,z,r,\phi) + \sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial}{\partial \phi} V(t,z,r,\phi)\right)\left(i\, r^2 + i\, |r+1||r-1| - i\right)}{8\,(r+1)(r-1) r^2 F(t)^2\, V(t,z,r,\phi)}$$

$$NP14 = \frac{\left(r^2 F(t)\, \sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial}{\partial t} F(t)\, \frac{\partial}{\partial t} V(t,z,r,\phi) + r^2 F(t)^2 \sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial^2}{(\partial t)^2} V(t,z,r,\phi) - r^2 |r+1||r-1| \frac{\partial}{\partial t} F(t)\, \frac{\partial}{\partial r} V(t,z,r,\phi) + r^4 \sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial^2}{(\partial r)^2} V(t,z,r,\phi) - rV(t,z,r\right.}{}$$

$$NP16 = -\frac{\left(rV(t,z,r,\phi)\, \sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial}{\partial t} F(t)^2 - rF(t)\, \sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial}{\partial t} F(t)\, \frac{\partial}{\partial t} V(t,z,r,\phi) - r|r+1||r-1| \frac{\partial}{\partial t} F(t)\, \frac{\partial}{\partial r} V(t,z,r,\phi) - 2\, r^3 \sqrt{|r+1|}\sqrt{|r-1|} \frac{\partial^2}{(\partial r)^2} V(t,z,r,\phi) + V(t,z,r,\phi\right.}{8\, rF(t)^2\, V(t,z,r,\phi)\, |r+1|^{\frac{3}{2}} |}$$

$NP17 =$

$$-\frac{\left(rV\left(t,z,r,\phi\right)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F\left(t\right)^{2}+rF\left(t\right)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F\left(t\right)\frac{\partial}{\partial t}V\left(t,z,r,\phi\right)+r|r+1||r-1|\frac{\partial}{\partial t}F\left(t\right)\frac{\partial}{\partial r}V\left(t,z,r,\phi\right)+2\,r^{3}\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^{2}}{(\partial r)^{2}}V\left(t,z,r,\phi\right)+V\left(t,z,r,\phi\right)\cdots}{8\,rF\left(t\right)^{2}V\left(t,z,r,\phi\right)|r+1|^{\frac{3}{2}}|\cdots}$$

$$NP18 = \frac{\left(rF\left(t\right)\frac{\partial^{2}}{\partial t\partial\phi}V\left(t,z,r,\phi\right)-r\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^{2}}{\partial r\partial\phi}V\left(t,z,r,\phi\right)-r\frac{\partial}{\partial t}F\left(t\right)\frac{\partial}{\partial\phi}V\left(t,z,r,\phi\right)+\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial\phi}V\left(t,z,r,\phi\right)\right)\left(i\,r^{2}+i\,|r+1||r-1|-i\right)}{8\,(r+1)(r-1)r^{2}F\left(t\right)^{2}V\left(t,z,r,\phi\right)}$$

Equations are vanished.

Bianchi Identities

$BI1$

$$=\frac{\sqrt{2}\Big(r^{3}F\left(t\right)^{2}V\left(t,z,r,\phi\right)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F\left(t\right)\frac{\partial^{2}}{\partial t\partial z}V\left(t,z,r,\phi\right)-r^{3}F\left(t\right)^{2}\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F\left(t\right)\frac{\partial}{\partial t}V\left(t,z,r,\phi\right)\frac{\partial}{\partial z}V\left(t,z,r,\phi\right)+i\,r^{2}F\left(t\right)^{2}V\left(t,z,r,\phi\right)^{2}\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^{3}}{(\partial t)^{2}\partial\phi}V\left(\cdots\right.}{\cdots}$$

$BI2$

$$=\frac{\sqrt{2}\Big(2\,r^{3}V\left(t,z,r,\phi\right)^{3}\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F\left(t\right)^{3}+r^{3}F\left(t\right)V\left(t,z,r,\phi\right)^{3}\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F\left(t\right)\frac{\partial^{2}}{(\partial t)^{2}}F\left(t\right)-2\,r^{3}F\left(t\right)^{2}V\left(t,z,r,\phi\right)^{3}\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^{3}}{(\partial t)^{3}}F\left(t\right)-3\,r^{3}F\left(t\right)V\left(t,z,r,\phi\right)^{2}\cdots}{\cdots}$$

$BI3 =$

$$-\frac{\sqrt{2}\Big(2\,r^{3}V\left(t,z,r,\phi\right)^{3}\frac{\partial}{\partial t}F\left(t\right)^{3}+r^{3}F\left(t\right)V\left(t,z,r,\phi\right)^{3}\frac{\partial}{\partial t}F\left(t\right)\frac{\partial^{2}}{(\partial t)^{2}}F\left(t\right)-2\,r^{3}F\left(t\right)^{2}V\left(t,z,r,\phi\right)^{3}\frac{\partial^{3}}{(\partial t)^{3}}F\left(t\right)-3\,r^{3}F\left(t\right)V\left(t,z,r,\phi\right)^{2}\frac{\partial}{\partial t}F\left(t\right)^{2}\frac{\partial}{\partial t}V\left(t,z,r,\phi\right)-3\,r^{3}F\left(t\right)^{2}V\left(t,z,r,\phi\right)^{2}\cdots}{\cdots}$$

$BI4 =$

$$-\frac{\sqrt{2}\Big(r^{3}F\left(t\right)^{2}V\left(t,z,r,\phi\right)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F\left(t\right)\frac{\partial^{2}}{\partial t\partial z}V\left(t,z,r,\phi\right)-r^{3}F\left(t\right)^{2}\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F\left(t\right)\frac{\partial}{\partial t}V\left(t,z,r,\phi\right)\frac{\partial}{\partial z}V\left(t,z,r,\phi\right)-i\,r^{2}F\left(t\right)^{2}V\left(t,z,r,\phi\right)^{2}\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^{3}}{(\partial t)^{2}\partial\phi}\cdots}{\cdots}$$

$BI5 =$

$$-\frac{\sqrt{2}\Big(2\,r^{3}V\left(t,z,r,\phi\right)^{3}\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F\left(t\right)^{3}-r^{3}F\left(t\right)V\left(t,z,r,\phi\right)^{3}\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F\left(t\right)\frac{\partial^{2}}{(\partial t)^{2}}F\left(t\right)+3\,r^{3}F\left(t\right)V\left(t,z,r,\phi\right)^{2}\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F\left(t\right)^{2}\frac{\partial}{\partial t}V\left(t,z,r,\phi\right)+r^{3}F\left(t\right)^{2}\cdots}{\cdots}$$

$$BI6 = -\frac{\sqrt{2}\Big(2\,r^3 V\left(t,z,r,\phi\right)^3 \sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F\left(t\right)^3 + 3\,r^3 F\left(t\right) V\left(t,z,r,\phi\right)^3 \sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F\left(t\right)\frac{\partial^2}{(\partial t)^2}F\left(t\right) - r^3 F\left(t\right) V\left(t,z,r,\phi\right)^2 \sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F\left(t\right)^2 \frac{\partial}{\partial t}V\left(t,z,r,\phi\right) + r^3 F\left(t\right)^2 \cdots}{\cdots}$$

$$BI7 = -\frac{\sqrt{2}\Big(r^2 F\left(t\right)^2 V\left(t,z,r,\phi\right)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^3}{(\partial t)^2\partial \phi}V\left(t,z,r,\phi\right) + 2\,r^2 F\left(t\right) V\left(t,z,r,\phi\right)|r+1||r-1|\frac{\partial^3}{\partial t\partial r\partial \phi}V\left(t,z,r,\phi\right) + r^2 F\left(t\right) V\left(t,z,r,\phi\right)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F\left(t\right)\frac{\partial^2}{\partial t\partial \phi}V\left(t,z,r,\phi\right) - \cdots}{\cdots}$$

$$BI8 = \frac{\sqrt{2}\Big(r^2 F\left(t\right)^2 V\left(t,z,r,\phi\right)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^3}{(\partial t)^2\partial \phi}V\left(t,z,r,\phi\right) + 2\,r^4 F\left(t\right) V\left(t,z,r,\phi\right)\frac{\partial^3}{\partial t\partial r\partial \phi}V\left(t,z,r,\phi\right) + 2\,r^2 F\left(t\right) V\left(t,z,r,\phi\right)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F\left(t\right)\frac{\partial^2}{\partial t\partial \phi}V\left(t,z,r,\phi\right) - r^2 F\left(t\right)^2 \sqrt{|r \cdots}{\cdots}$$

$$BI9 = -\frac{\sqrt{2}\Big(r^3 V\left(t,z,r,\phi\right)^2 \sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F\left(t\right)^3 + 3\,r^3 F\left(t\right) V\left(t,z,r,\phi\right)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F\left(t\right)^2 \frac{\partial}{\partial t}V\left(t,z,r,\phi\right) - 3\,r^3 F\left(t\right)^2 V\left(t,z,r,\phi\right)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F\left(t\right)\frac{\partial^2}{(\partial t)^2}V\left(t,z,r,\phi\right) - r^3 \cdots}{\cdots}$$

$$BI10 = \frac{\sqrt{2}\Big(r^3 F\left(t\right)^2 V\left(t,z,r,\phi\right)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^3}{\partial t\partial z\partial r}V\left(t,z,r,\phi\right) - r^3 F\left(t\right)^2 \sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^2}{\partial t\partial r}V\left(t,z,r,\phi\right)\frac{\partial}{\partial z}V\left(t,z,r,\phi\right) - r^3 F\left(t\right) V\left(t,z,r,\phi\right)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F\left(t\right)\frac{\partial^2}{\partial z\partial r}V\left(t,z,r,\phi\right) \cdots}{\cdots}$$

$$BI11 = -\frac{\sqrt{2}\Big(r^3 V\left(t,z,r,\phi\right)^2 \sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F\left(t\right)^3 + 2\,r^3 F\left(t\right) V\left(t,z,r,\phi\right)^2 \sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F\left(t\right)\frac{\partial^2}{(\partial t)^2}F\left(t\right) + r^3 F\left(t\right) V\left(t,z,r,\phi\right)\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F\left(t\right)^2 \frac{\partial}{\partial t}V\left(t,z,r,\phi\right) - 3\,r^3 F\left(t\right)^2 V \cdots}{\cdots}$$

Equations are not vanished.

Petrov Type

Type "I"